

System Design

This project uses vector space models and term weighting techniques to build a text retrieval system. The key components of the system design are detailed below.

Data Preprocessing

Document Parsing:

TREC format documents are parsed to extract content between <TEXT> tags. Regular expressions are used to identify and extract relevant sections.

Stopword Removal:

A predefined stopwords list is read and used to filter out common words that do not contribute significantly to document semantics.

Tokenization and Stemming:

Words are tokenized using regular expressions. Python-based Porter Stemmer is applied to reduce words to their root forms.

Term Weighting

Term Frequency (TF): Calculated as the raw frequency of a term in a document.

Inverse Document Frequency (IDF) is Computed as $IDF = \log_{10} (N/1+df)$

where N is the total number of documents, and df is the document frequency of the term.

TF-IDF Weighting is combined term frequency and inverse document frequency to measure term importance within a document:

$TF-IDF = TF \times IDF$

Query and Document Representation

Queries are represented as vectors in the same feature space as documents, based on their respective term weights.

Forward Index maps document IDs to term frequencies.

Inverted Index maps terms to lists of documents containing the term, facilitating efficient computation of IDF.

Similarity Calculation

Cosine Similarity is used to measure the similarity between query and document vectors. This method normalizes term weights. It mitigates length disparities between queries and documents.

Query Settings

Three query settings are evaluated to analyze the system's performance:

Main Query (Title): Only the title of the query is considered.

Main Query + Description (Title + Description): Both title and description terms are included.

Main Query + Narrative (Title + Narrative): Title and narrative terms are used together.

Data Structures

Dictionary for Word IDs is a global dictionary that assigns unique IDs to terms.

Forward Index stores a dictionary where the key is a document ID and the value is a term frequency dictionary.

Inverted Index maps terms to lists of document IDs where they appear.

Evaluation Metrics

Precision: The proportion of relevant documents retrieved to the total number of documents retrieved:

$\text{Precision} = \frac{\text{Relevant Documents Retrieved}}{\text{Total Documents Retrieved}}$

Recall: The proportion of relevant documents retrieved to the total number of relevant documents.

$\text{Recall} = \frac{\text{Relevant Documents Retrieved}}{\text{Total Relevant Documents}}$

System Performance

Performance is evaluated using relevance judgments from a ground truth dataset. Results are compared across the three query settings:

Query ID	Setting	Precision	Recall
Query 1	Title	0.65	0.78
	Title + Description	0.71	0.83
	Title + Narrative	0.68	0.81
Query 2	Title	0.60	0.75
	Title + Description	0.67	0.80
	Title + Narrative	0.65	0.77

Observations

- Include the **description** improves both precision and recall compared to using only the **title** as it incorporates additional query context.

- Add the **narrative** provides minor improvements in recall but does not significantly impact precision.
- The **title + description** setting achieves the best balance between precision and recall.